Your new `Calculation_main.tex`: check, and consistency with Group I, II, III

1 Answer

- **Your Group I, II, III cross sections agree with** `Calculation_main.tex`, once the corrections listed earlier (`groups23/main/Main_corrections.pdf`) are made. Your Group files already use the correct forms everywhere they differ from what is written in `Calculation_main.tex` (Sec. 3).
- **In the new file, two of the earlier corrections are done:**
  - E9, `stseve` [line 355]: the energy denominators are now $\left(\frac{\mathbf{p}^2}{2p^+}\right)\left(\frac{\mathbf{r}^2}{2r^+}+\frac{\mathbf{q}^2}{2q^+}\right)\left(\frac{\mathbf{k}^2}{2k^+}\right)$. Correct.
  - M8 [line 398]: "Setting $\Lambda = 0$, wherever possible" → "Simplifying". Correct: nothing is set to zero in the next line.

  Apart from these two lines the file is identical to the one checked before.
- **All other corrections are still needed** (Sec. 2). Seven of them change results if `Calculation_main.tex` is used as written (E1–E7). Your Group files are not affected by them, because the Group files were built from the correct forms.
- **Remove the text after** `\end{document}` (lines 1965–2031): it is a copy of your message to me. LaTeX ignores it, but it should not be in the file.

2 What is still to be corrected in `Calculation_main.tex`

Line numbers are those of your new file (the same as before). The marked-up copy is `Calculation_main_corrected.pdf` (red: as written, blue: corrected).

	where	what	your Group files
Errors that change results			
E1	`onegrho12` [683]	overall sign: $-\frac{ig^2 f^{abc}}{8\pi^3} \to +\frac{ig^2 f^{abc}}{8\pi^3}$; its ξ-form below is right	Group I Row III uses the right sign
E2	`Eq:onegrhoX1` [701]	overall sign: $+\frac{ig^2 f^{abc}}{4\pi^3} \to -\frac{ig^2 f^{abc}}{4\pi^3}$	Group I Row III uses the right sign
E3	$\mathbb{B}_3$ [1276–1300]	the non-instantaneous $\Theta(p^+ - k^+ - \Lambda)$ entry needs a minus sign; then every $1/\Lambda$ term cancels	right in Group I Row III: its $\frac{1}{\Lambda}$ lines add up to zero (3×10^{-16})
E4	$\mathbb{A}_2^{(3)}$, momentum space [1252]	counts $\frac{\pi^2}{3}-\frac{67}{9}$ twice and misses the pole of Ψ^B; use the bracket of `grhoposition`	Group I Row II part 1 uses `grhoposition` (right)
E5	unitarity, $\mathbb{A}_1^{(3)\dagger}$ [1087–1101]	$\mathbb{N}^{(2)}\mathbb{A}^{(1)} = \mathbb{A}^{(1)}\mathbb{N}^{(2)}$ is false; keep $-[\mathbb{N}^{(2)}, \mathbb{A}_2^{(1)}]$	its commutator is a two-ρ log term of Group I Row II; with it the two-ρ sum is JIMWLK⊗LO
E6	$\mathbb{B}_2$, (k^+, p^+) forms [1260, 1268]	$\Theta(k^+ - \Lambda - p^+) \to \Theta(p^+ - \Lambda)\Theta(k^+ - \Lambda)\Theta(\vee - k^+ - p^+)$	Groups II, III use $\Lambda < p^+ < \vee - k^+$ (right)
E7	Appendix F [1857–1922]	JIMWLK formula: M at the points of $\mathcal{K}(u,v,t)$; $dN^{(5)}$ with $+$	M2′ of `Checks_JIMWLK_BFKL.pdf`
Typos (the next line is right)			
E8	`part2` [270]	$\delta^{ni} \to \delta^{ji}$	not used
E10	`onegrho22` [731]	"+" → product	not used
E11	`A1coeff` [1247]	$\mathbb{A}^a_{2i}(k^+, \mathbf{z}) \to \mathbb{A}^a_{2i}(k^+, -\mathbf{z})$	right in the Groups
Exact forms (replace every $\Lambda \to 0$, $\vee \to \infty$ step; only $\vee > p^+, k^+ > \Lambda$)			
X1–X4	`gtwoloop1`, `instgg`, `onerho2`, `tworho2`, `combinedgrhorho`	exact ξ integrals, $\ln\frac{\Lambda}{k^+} \to \ln\frac{\Lambda}{k^+-\Lambda}$	the Groups keep Λ, $\vee$ exact
X5	`threerhoafter` [411], after `zetainteg` [432]	"Using the argument when $\vee$ is very large or Λ is very small": not needed, the $\frac{1}{\Lambda}-\frac{1}{\vee}$ terms cancel exactly (your M8 fix is the first part of X5)	three-ρ terms

X6, X7	`grhoposition`, `combinedgrhorho(pos)` [496–540]	exact ξ limits $[\frac{k^+}{\vee}, \frac{k^+}{k^+ + \Lambda}]$; keep Λ	Group I Row II part 1 has $\log(1 - \frac{k^+}{\vee})$: exact
Only for a general adjoint U (with $U^{bc} = U^{cb}$ written = corrected)			
G1–G4	$\mathbb{X}_1$, `eq: A1a`, $\bar{\mathbb{C}}\mathbb{A}$, `eq:CS2`	index placements	as in your convention
Minor			
M1–M7		notation, typography, order-g^4 normalisation ($\frac{3}{8}$, not $-\frac{1}{8}$; not used)	–

Programmatically (`pending.py`): the "as written" text of every item above is still in your file; only E9 and the M8 sentence are no longer found.

3 `Calculation_main.tex` → your Group I, II, III

The chain of checks:

1. **Wave functions and coefficients** of `Calculation_main.tex` (Ψ's, $\mathbb{A}^{(1)}$, $\mathbb{A}^{(3)}$, $\mathbb{B}_2$, $\mathbb{B}_3$, $\mathbb{C}_1$, $\mathbb{X}_{1,2}$, Eq. (CS2)) were rebuilt from the matrix elements m1–m5 and compared point by point (`groups23/main`). The results are the corrected forms of Sec. 2.

2. **The first lines of every row** of your Group files (the products of these coefficients, before any p^+, w, w' integral) were compared numerically with that first-principles evaluation: Group I in `group1` (A3, B3, B2, C1 and the cross-section formula), Groups II and III in `groups23` (random points, random adjoint Wilson lines, random charges; exact p^+ integrals).

3. **Your current Group files have the same first lines.** Equation by equation (`eq_diff.py`; colours, spaces and the lists of points under $\int$ ignored):

Eq.	lines	against the version checked against `Calculation_main.tex`
Group I (checked: `group1/input/Group_I_v2.tex`)		
(unnumbered)	41–43	identical
(1.1)	45–47	rewritten; checked step by step (item 4 below)
(1.2)	49–51	rewritten; checked step by step (item 4 below)
(1.3)	55–57	identical
(1.4)	58–60	identical
(1.5)	68–73	rewritten; checked step by step (item 4 below)
(1.6)	74–79	identical
(1.7)	81–86	identical
(unnumbered)	88–90	identical
(1.8)	91–93	identical
(1.9)	95–97	rewritten; checked step by step (item 4 below)
(1.10)	98–100	rewritten; checked step by step (item 4 below)
(1.11)	101–120	rewritten; checked step by step (item 4 below)
(1.12)	121–129	identical
(unnumbered)	131–133	identical
(unnumbered)	135–137	identical
(1.13)	138–140	identical
(1.14)	142–145	rewritten; checked step by step (item 4 below)
(unnumbered)	147–149	identical
(1.15)	152–154	identical
(1.16)	156–158	rewritten; checked step by step (item 4 below)
(1.17)	160–162	rewritten; checked step by step (item 4 below)
(1.18)	164–166	identical
(1.19)	168–170	rewritten; checked step by step (item 4 below)
(1.20)	172–174	rewritten; checked step by step (item 4 below)
(1.21)	177–179	rewritten; checked step by step (item 4 below)
(1.22)	181–183	rewritten; checked step by step (item 4 below)
(1.23)	184–186	rewritten; checked step by step (item 4 below)
(1.24)	188–190	rewritten; checked step by step (item 4 below)
(1.25)	193–198	identical
(1.26)	200–204	rewritten; checked step by step (item 4 below)
(1.27)	205–218	identical
(1.28)	220–232	rewritten; checked step by step (item 4 below)
(1.29)	234–241	rewritten; checked step by step (item 4 below)

(1.30)	244–246	rewritten; checked step by step (item 4 below)
(1.31)	248–250	rewritten; checked step by step (item 4 below)
(1.32)	251–255	rewritten; checked step by step (item 4 below)
(1.33)	256–263	rewritten; checked step by step (item 4 below)
Group II (checked: `groups23/input/Group_II.tex`)		
(unnumbered)	39–41	identical
(1.1)	45–47	identical
(1.2)	49–51	rewritten; checked step by step (item 4 below)
(1.3)	53–55	identical up to the measure
(1.4)	57–59	identical up to the measure
(1.5)	61–63	identical up to the measure
(1.6)	65–67	identical
(1.7)	68–70	rewritten; checked step by step (item 4 below)
(1.8)	71–73	identical
(1.9)	74–76	identical
(1.10)	77–79	identical
(1.11)	80–82	identical
(unnumbered)	88–90	identical
(1.12)	91–95	identical up to the measure
(1.13)	97–101	identical up to the measure
(1.14)	103–105	identical up to the measure
(1.15)	107–109	rewritten; checked step by step (item 4 below)
(1.16)	111–113	rewritten; checked step by step (item 4 below)
(1.17)	115–117	rewritten; checked step by step (item 4 below)
(1.18)	119–121	rewritten; checked step by step (item 4 below)
(1.19)	123–125	rewritten; checked step by step (item 4 below)
(1.20)	127–129	rewritten; checked step by step (item 4 below)
(1.21)	131–133	rewritten; checked step by step (item 4 below)
Group III (checked: `groups23/input/Group_III.tex`)		
(unnumbered)	39–42	identical
(1.1)	43–45	identical up to the measure
(1.2)	47–49	rewritten; checked step by step (item 4 below)
(unnumbered)	51–53	identical
(1.3)	54–56	identical up to the measure
(1.4)	58–60	rewritten; checked step by step (item 4 below)
(unnumbered)	62–64	identical
(1.5)	67–69	identical up to the overall sign (**M1** of the Group II/III check, the fix you made)
(1.6)	71–73	identical up to the overall sign (**M1** of the Group II/III check, the fix you made)
(1.7)	75–77	rewritten; checked step by step (item 4 below)
(1.8)	79–82	identical up to the overall sign (**M1** of the Group II/III check, the fix you made)
(1.9)	84–87	identical up to the overall sign (**M1** of the Group II/III check, the fix you made)
(1.10)	89–93	rewritten; checked step by step (item 4 below)
(unnumbered)	97–99	identical up to the overall sign (**M1** of the Group II/III check, the fix you made)
(1.11)	100–102	identical up to the overall sign (**M1** of the Group II/III check, the fix you made)
(1.12)	104–106	identical up to the overall sign (**M1** of the Group II/III check, the fix you made)
(1.13)	108–110	rewritten; checked step by step (item 4 below)
(1.14)	112–114	rewritten; checked step by step (item 4 below)
(unnumbered)	116–118	identical up to the overall sign (**M1** of the Group II/III check, the fix you made)
(1.15)	119–121	identical up to the overall sign (**M1** of the Group II/III check, the fix you made) up to the measure
(1.16)	123–125	identical up to the overall sign (**M1** of the Group II/III check, the fix you made)
(1.17)	127–129	rewritten; checked step by step (item 4 below)

4. **Every later step** of your current files (the rewritten equations above included) was checked numerically, from one equation to the next (`full_check/step_check.py`, `Checks_JIMWLK_BFKL.pdf`, Sec. 2): all agree to 10^{-16}–6×10^{-12} (plus prescriptions up to $O(\Lambda)$), apart from the typo M2 (Group II Eq. (1.16)). The changed equations of Group II Row IV (1.15)–(1.21) are the $d_\perp$-general form (`groups23/rowIV_d`); those of Group I are the log rewritings and E1.

So the Group files follow from `Calculation_main.tex` with its corrections. Group III Row III's first unnumbered line still has the old sign (M3 of `Checks_JIMWLK_BFKL.pdf`); the explicit lines below it are right.

4 Files

- `Calculation_main_corrected.pdf`: your new file with every remaining correction marked in place (red/blue); its source `Calculation_main_corrected.tex` keeps your preamble (revtex4-2, tikzit); the PDF is built from a copy with the article class and the diagrams as boxes.

- `pending.py` → `pending.json`; `eq_diff.py` → `eq_diff.json`.
- The explanations of E1–E11, X1–X7, G1–G4: `groups23/main/Main_corrections.pdf`.